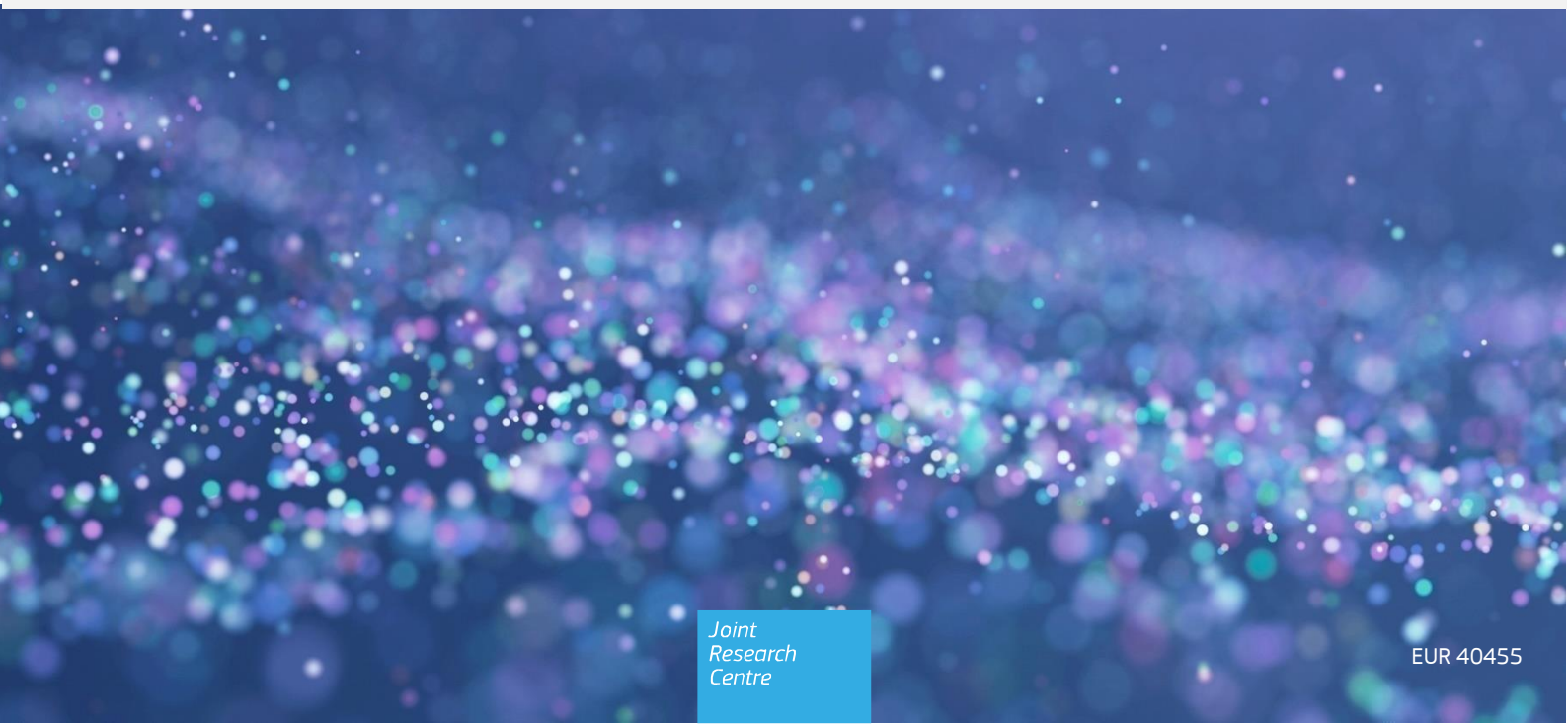
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Energy Justice for AI: A Framework

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Abstract

The deployment of AI technologies is expected to generate substantial economic and societal improvements in the energy sector and is already playing a pivotal role in optimizing operations, enhancing efficiency, aiding decarbonisation, reducing costs, enhancing system resilience and driving innovation. In this way, AI could help to promote a just and fair transition, yet, concerns remain in relation to its potential impacts on safety, privacy, transparency, fairness or accountability. The recent EU AI Act seeks to protect fundamental rights, promote ethical development and use of AI and ensure accountability and transparency in AI systems. It thus has implications for a just transition. In this regard, the AI Act requires that systems it classifies as "high risk" (including energy infrastructure such as smart grids), should undergo a risk assessment. However, no EU-level framework for energy justice exists and existing "responsible AI" frameworks may not fully capture the energy-specific context and the complexity of the relationship between energy and society. An energy justice framework for AI can help address these gaps by providing a more nuanced understanding of the energy sector's unique challenges and opportunities. It can also help ensure that AI development in the energy sector acknowledges the needs and rights of disadvantaged, underserved or marginalized communities, promotes energy democracy, and supports a just transition to a low-carbon economy. In this report we first explore the potential energy justice implications of the use of AI in the energy sector. We then propose a tentative energy justice assessment framework tailored to AI applications in the energy sector. We use in-house (JRC) examples of use-cases of AI in the energy transition to demonstrate the framework. Such a guidance framework can help to check whether energy justice is likely to be promoted by a given AI application.

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1. Introduction

1.1. Policy context and background

Energy justice is a policy priority in the EU. In 2019 the EU Green Deal established the goal that no one should be left behind and pointed out that a "transition can only succeed if it is conducted in a fair and inclusive way"[1]. In 2020 the EU communication *Shaping Europe's Digital Future* highlighted the twin challenge of a green and digital transformation, and the necessity for every citizen to have "a fair chance [...] to reap the benefits of our increasingly digitised society"[2]. The commitment to a just transition is restated in the 2025 EU Clean Industrial Deal that aims to deliver quality jobs, empower people by building on their skills, and promote social cohesion and equity across all regions[3]. With the upcoming Citizen Energy Package, the Commission intends "to make the Just Transition a reality, addressing energy poverty and supporting the decarbonisation of coal regions in transition"[4].

The energy transition involves shifting from fossil fuels toward renewable energy sources, greater electrification across various sectors, and the development of more efficient and resilient energy systems by integrating digital technologies into the energy sector. The EU Energy Digitalisation Action Plan[5] supports this digital transformation. Using digital technologies, particularly Artificial Intelligence (AI), is seen as essential to managing the complexities of the energy transition. The deployment of AI technologies is projected to generate substantial economic and societal improvements in various fields, such as environmental protection, healthcare, public administration, finance, transportation, home affairs, and agriculture (Madiega & Chahri, 2024). In the energy sector, AI is well positioned to facilitate the simultaneous expansion of smart grids and the vast volumes of data they generate (IEA, 2023) and is already playing a pivotal role in optimizing operations, enhancing efficiency, reducing costs, enhancing system resilience and driving innovation.

An IEA survey of grid operators from 13 countries, comprising Australia, Belgium, the People's Republic of China, Czechia, France, Germany, Ireland, Italy, Japan, Korea, Malaysia, the Netherlands and the United States, examined the use of AI application in electricity networks (Figure 1) and shows that AI supports electricity grids through the optimisation of operation planning and maintenance, including fault prevention by automated image recognition (IEA, 2025).

1 https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en

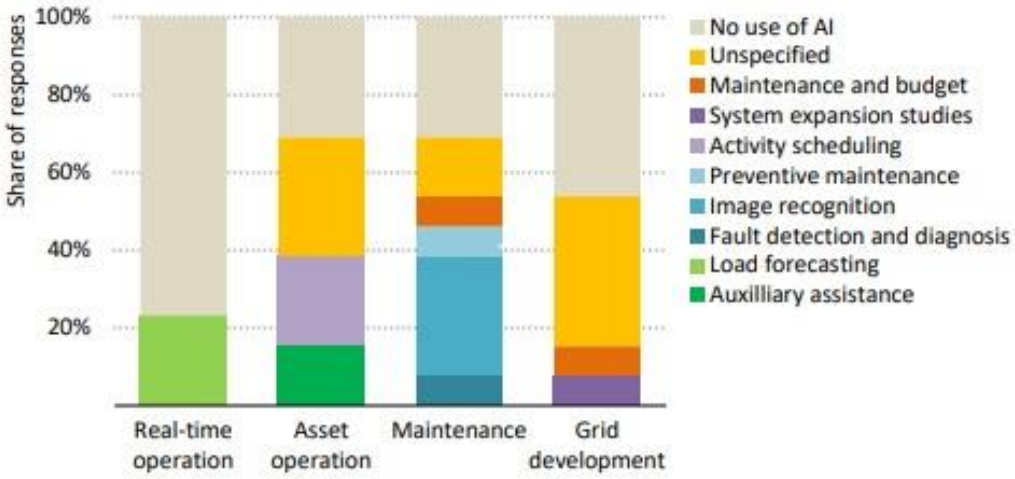
2 https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/shaping-europes-digital-future_en

3 https://commission.europa.eu/topics/eu-competitiveness/clean-industrial-deal_en

4 https://energy.ec.europa.eu/news/citizens-energy-package-commission-starts-consultation-process-2025-06-19_en

5 https://energy.ec.europa.eu/topics/eus-energy-system/digitalisation-energy-system_en

Figure 1. Electricity utilities using AI applications by categories, 2024



Source: IEA, 2025

Therefore, AI could be instrumental for a just and fair transition, e.g. by empowering consumers via increased transparency in pricing, maintaining the affordability of electricity in the energy transition or helping the EU reach its policy targets e.g. on decarbonisation or energy efficiency. At the same time, the integration of AI into energy systems has caused concern in relation to its potential negative social impacts. Not only could AI introduce new policy challenges, it could also perpetuate or indeed intensify existing impacts of digitalisation of the energy sector, e.g. by reducing safety, privacy, transparency, fairness or accountability, among others (Noorman et al., 2023), potentially making existing gaps wider and burdens heavier for disadvantaged or vulnerable groups. A recent JRC report further investigates the technological, economic, and societal challenges of generative AI, with the goal of supporting policymakers in navigating the opportunities and challenges presented by this technology (Abendroth Dias et al., 2025).

1.2. The EU AI Act and the energy sector

The European Union has made AI regulation a key policy priority in recent years. Aiming for a 'human-centric' approach rooted in EU values, policymakers moved from non-binding guidelines in 2019 to a legislative path, culminating in the landmark AI Act signed in June 2024. This first-of-its-kind global binding regulation establishes a unified framework for AI within the EU (Madiega & Chahri, 2024). The AI Act generally seeks to protect fundamental rights, promote ethical development and use of AI and ensure accountability and transparency in AI systems. The EU AI Act has therefore implications for energy justice. By embedding these principles into its regulatory framework, the EU aims to create a legal environment that fosters responsible AI innovation.

Utilizing a 'risk-based approach', the AI Act assigns different levels of regulation to AI systems based on their potential for harm. Systems presenting 'unacceptable' risks are prohibited. 'High-risk' AI, which could endanger health, safety, or fundamental rights, is authorized but subject to stringent requirements for market access. AI with transparency concerns faces information obligations, while minimal-risk AI is exempt from further regulation. This risk assessment is determined by the AI's intended purpose and function, particularly in specified high-risk use cases, including safety components of EU-regulated products and applications listed in a Commission-managed database.

(Madiega & Chahri, 2024). High-risk systems (according to Annex III[6]) include those that impact critical infrastructure such as AI systems intended to be used as safety components in the management and operation of critical digital infrastructure, road traffic, or in the supply of water, gas, heating or electricity. The regulation also establishes specific rules for general-purpose AI (GPAI) models, with stricter requirements for those with 'high-impact capabilities' that could create systemic risks and significantly affect the EU's internal market (Madiega & Chahri, 2024). This is relevant for the energy sector as well, because many grid management systems will fall under the high-risk category[7]. Therefore, actors in the electricity sector that develop or use AI systems that are classified as high-risk must ensure that those systems meet the requirements introduced by the AI Act for high-risk AI systems and must comply with specific obligations. Furthermore, the AI Act requires that high-risk AI systems have effective human oversight to prevent or mitigate risks from automation failures. This implies that DSOs and TSOs must define the roles and responsibilities of human operators who can intervene in AI decision-making, ensuring personnel are trained to carry out these responsibilities. This is to ensure that humans remain in control, and that AI systems are not allowed to operate in ways that are harmful or unethical.

Public sector bodies and entities providing public services are required to conduct fundamental rights impact assessments before deploying high-risk AI systems. This requirement emphasizes the importance of proactively considering the potential impact of AI on individuals' rights.

1.3. The need for an energy justice approach

As we have introduced, integrating AI into energy systems creates opportunities and challenges for society which must be recognised and managed to ensure a just energy transition. Several frameworks to promote the socially responsible use of AI exist, such as the European Union's High-Level Expert Group on Artificial Intelligence (AI HLEG) guidelines (European Commission, 2020) or the IEEE's Ethics of Autonomous and Intelligent Systems (IEEE, 2025). However, notwithstanding the existence of these frameworks, there is a need for an energy justice framework to guide AI development in the energy sector. EU legislation has been criticised for inadequate attention to energy justice principles, despite an urgent need for them (EERA, 2023). Energy justice frameworks prioritize the unique challenges and opportunities presented by the energy sector. The existing “responsible AI” frameworks may not fully capture the energy-specific context and the complexity of the relationship between energy and society. An energy justice framework can help address these gaps by providing a more nuanced understanding of the energy sector's unique challenges and opportunities. It can also help ensure that AI development in the energy sector prioritizes the needs and rights of disadvantaged, underserved or marginalized communities, promotes energy democracy, and supports a just transition to a low-carbon economy.

Furthermore, how social trust will evolve will depend on AI implementation across all sectors, not just within energy, which will together shape the broader perception of AI reliability and value (IEA, 2025). It is possible that the expansion of AI, i.e. transitioning from digitalisation to AI-driven intelligent systems, may result in a reduction of social trust, especially regarding applications involving safety risks (e.g. autonomous vehicles) or potential impact on finances (e.g. building energy management).

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6 https://artificialintelligenceact.eu/annex/3/

7 High-Risk AI System: an AI system that poses a significant risk to the health, safety, or fundamental rights of individuals, as defined by the EU AI Act, and subject to specific compliance requirement

A recent study highlighted how even in countries where trust has been a foundational cultural value, trust has been eroded via the increased use of big data. Social trust could be further eroded if the choice of individual consumers to opt in or out is not consistently given as AI becomes more integrated and pervasive in the daily life (Godoy et al., 2021).

Concerns have also arisen, for instance, around the private sector push for a 10-year ban on US states regulating AI (Financial Times, 2025) and the fact that big tech has already heavily influenced and some would argue weakened the Code of Practice for general-purpose AI models (Corporate Europe, 2025), a key implementation tool for the AI Act. Hence, in a context where technological development is outpacing ethical and legal safeguards, trust will be more important than ever for the success of the (digital) energy transition. It is therefore imperative to address any potential concerns and ensure transparency, accountability and human autonomy. Using an energy justice framework can also help in this regard.

1.4. Purpose of this report

As previously discussed, no formal guidelines exist at EU level for assessing the energy justice implications of AI. In this report we first explain in detail the potential energy justice implications of the use of AI in the energy sector. Based on a review of existing energy justice literature and a preliminary exploration of the potential energy justice implications of AI, we propose a tentative energy justice framework for assessing the risks of AI applications in the energy sector (Section 3.5). We stress that the framework is non-exhaustive and should be open to change. We also use in-house (JRC) examples of use-cases of AI in the energy transition to demonstrate the framework (Section 4). Such a guidance framework can help to check whether energy justice is likely to be promoted by a given AI application in the energy sector.

2. Methodology

In order to design a framework tailored to measure the energy justice implication of AI we carried out the following:

- Synthesis of existing energy justice literature: using the current literature on energy justice and our previous work of a similar nature (Shortall & Mengolini, 2024), we created a synthesis of the general energy justice concept (Table 1).
- Tailoring of the energy justice concept to AI: for each of the dimensions of energy justice in Table 1, we searched for relevant exemplary literature about the use of AI in the energy sector. This was not an exhaustive or systematic literature review, but rather a preliminary exploration of the potential energy justice implications of AI. We also added a fourth dimension, data justice, to the energy justice framework, due to its importance in the context of AI and the fact that the importance of data justice is overarching for each of the dimensions of energy justice.
- Apply the framework: as a demonstration of the framework, we analysed JRC in-house examples of use-cases of AI in the energy transition using the four categories (and their sub-categories) of our tailored framework. We picked recent examples that illustrate common applications of AI in the energy sectors. This served as a demonstration of the use of the framework as well as a way to check for any obvious omissions.

3. Energy justice and AI

A just energy transition should be one that eliminates existing negative impacts and does not create new ones (Miller et al., 2013). For this reason, it is important to assess the justice implications of energy policies in all their dimensions. Using the lens of energy justice to guide policy formulation and assessment may allow us to identify critical parts of the system where energy injustices may occur and ensure that no aspects of the just transition are disregarded (EERA, 2023). In this section we examine how existing energy justice frameworks may be tailored to reflect AI concerns, by considering some of the energy justice implications of using AI in energy.

Energy justice is a comprehensive, people-focused framework that tackles unfairness and disparities in the energy industry. By acknowledging the importance of inclusive decision-making and considering the consequences of energy systems on various individuals and groups, it strives to create a more equitable energy system. Traditionally, the energy system has been viewed through a technological and economic lens, but energy justice introduces a human-centred perspective. It examines how the principles of justice may inform decision-making about energy production, distribution, consumption and its effects (Sovacool et al., 2016).

In the literature, several frameworks for energy justice exist and there is a degree of overlap and mutual reinforcement between all of them. Having the greatest uptake to date, McCauley's framework (Mccauley et al., 2013), with its roots in environmental and climate justice but with a focus on energy, has the aim of providing universal, safe, affordable and sustainable energy. It comprises three dimensions of energy justice (Table 1): distributional justice, justice as recognition, and procedural justice.

In addition, the overarching importance of cosmopolitan and restorative justice has also been recognised as important for energy justice (Heffron, 2024) as well as the need for consideration of the spatial and temporal nature of impacts (Heffron & McCauley, 2018). Cosmopolitan justice suggests that principles of justice should apply to all humans in all nations, and that our responsibilities do not stop at borders. Hence there is a need to consider the cross-border and regional effects of any policies (McCauley et al., 2019). Rather than being a fourth tenet of energy justice, some scholars argue that cosmopolitan justice is instead a cross-cutting issue (Van Uffelen et al., 2024) or applicable to the entire energy life-cycle (Heffron & McCauley, 2017). Similarly, restorative justice refers to the need to rectify any injustice, intended or unintended, caused at any point in energy decision-making (Heffron & McCauley, 2017). Hence there is a need for identifying foreseeable impacts in advance of policies (Heffron, 2024).

Building on McCauley's framework, Sovacool and Dworkin (Sovacool & Dworkin, 2015)[8], developed an 8 principle decision-making framework which is based on diverse philosophical concepts: virtue, utility, human rights, procedural justice, welfare, freedom, posterity, and responsibility. They propose a decision-making list of eight categories of assessment for energy justice decision-making: availability, affordability, due process, good governance, sustainability (resource depletion), intergenerational equity, intragenerational equity and responsibility. However, although intended to be a practical decision-making too, since it draws on an wide range of moral theories and viewpoints,

[8] This was updated in 2017 (Sovacool et al., 2017)

this framework may be more difficult to operationalise than the simpler three tenet framework (Velasco-Herrejon & Bauwens, 2020).

The aforementioned frameworks have also been criticised for being limited in how they account for unequal power relations and responsibilities in the energy system (Groves et al., 2021; Middlemiss et al., 2019), which is a particularly important consideration for energy poor/vulnerable groups. Including the normative goal of 'energy democracy' while assessing energy justice may be useful in directing energy transitions towards a more balanced distribution of power. While not explicitly defined, in practical terms it refers to an increase in empowerment of the citizen. The role of prosumers, co-operatives or municipalities replaces the role of energy companies, i.e. there is a fairer redistribution of power among energy system actors, and greater participation of citizens (Szulecki, 2018).

In Table 1 we provide a synthesis of the above-mentioned frameworks, in which we keep the three original dimensions from (McCauley et al., 2013) and incorporate aspects of the other frameworks where they overlap with these dimensions. For further discussion of these frameworks and their shortcomings, please see our previous report about energy justice assessment in the EU (Shortall & Mengolini, 2024).

Table 1. Synthesis of energy justice frameworks

Energy justice dimension	**Explanation**
Distributive justice	*Refers to:* where injustices occur, i.e. the distribution of benefits, harms or responsibilities due to developments in the energy system. *Calls for:* (environmental) burdens and benefits to be fairly distributed across society, borders and generations. Any injustices should be rectified
Justice as recognition	*Refers to*: who might be ignored, i.e. the fair representation of individuals within the energy system, as well as their right to political participation taking account of divergent perspectives due to culture, gender, ethnicity, etc. *Calls for*: characteristics and needs of a community to be recognised and individuals included, un-coerced and empowered in decision-making
Procedural justice	*Refers to*: whether or not there is fair (due) process, that engages all stakeholders in a non-discriminatory way *Calls for*: decision-making processes to be equitable, transparent and of high quality

Source: JRC, 2025

Many of the elements of existing energy justice frameworks resonate with existing ethical guidelines for AI—one of these is the European Union's High-Level Expert Group on Artificial Intelligence (AI HLEG) guidelines. These guidelines put forward a set of 7 key requirements that AI systems should meet to be deemed trustworthy. Based on these principles, the AI HLEG produced ALTAI (European Commission, 2020), a practical tool that translates the Ethics Guidelines into an accessible and

dynamic (self-assessment) checklist. The checklist can be used by developers and deployers of AI who want to implement the key requirements in practice[9].

The requirements include:

- Human agency and oversight: AI systems should empower human beings, allowing them to make informed decisions and fostering their fundamental rights. At the same time, proper oversight mechanisms need to be ensured, which can be achieved through human-in-the-loop, human-on-the-loop, and human-in-command approaches.
- Technical robustness and safety: AI systems need to be resilient and secure. They need to be safe, ensuring a fall-back plan in case something goes wrong, as well as being accurate, reliable and reproducible. That is the only way to ensure that unintentional harm can be minimized and prevented.
- Privacy and data governance: besides ensuring full respect for privacy and data protection, adequate data governance mechanisms must also be ensured, taking into account the quality and integrity of the data, and ensuring legitimised access to data.
- Transparency: data, system and AI business models should be transparent. Traceability mechanisms can help to achieve this. Moreover, AI systems and their decisions should be explained in a manner adapted to the stakeholder concerned. Humans need to be aware that they are interacting with an AI system and must be informed of the system's capabilities and limitations.
- Diversity, non-discrimination and fairness: unfair bias must be avoided, as it could have multiple negative implications, from the marginalization of vulnerable groups to the exacerbation of prejudice and discrimination. Fostering diversity, AI systems should be accessible to all, regardless of any disability, and involve relevant stakeholders throughout their entire life circle.
- Societal and environmental well-being: AI systems should benefit all human beings, including future generations. It must hence be ensured that they are sustainable and environmentally friendly. Moreover, they should take into account the environment, including other living beings, and their social and societal impact should be carefully considered.
- Accountability: mechanisms should be put in place to ensure responsibility and accountability for AI systems and their outcomes. Auditability, which enables the assessment of algorithms, data and design processes plays a key role therein, especially in critical applications. Moreover, adequate and accessible redress should be ensured.

Other frameworks to promote the socially responsible use of AI include the IEEE's Ethics of Autonomous and Intelligent Systems (IEEE, 2025), which provides general guidelines for the development and deployment of AI systems. Such frameworks also focus on principles like transparency, accountability, fairness, and human rights. Furthermore, the AI Act itself calls for the respect for fundamental rights in the development and use of AI systems. The connection with and the potential of fundamental rights to enhance energy justice has been highlighted by (Basil & Heffron, 2025). The fundamental rights (European Parliament, 2025) are respect for human dignity, freedom, democracy, equality, the rule of law, and respect for human rights, including the rights of

[9] https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai

persons belonging to minorities. This includes the protection of personal data: in a digital society where personal data are constantly collected, used and distributed, citizens should be able to decide freely how to use their own personal data to avoid abuse. Article 8 of the Charter of Fundamental Rights of the European Union[10] provides the right for everyone to the protection of personal data concerning them. Such data must be processed fairly for specified purposes and on the basis of the consent of the person concerned or some other legitimate basis laid down by law. Everyone has the right of access to data which has been collected concerning them, and the right to have it rectified. We elaborate further on concerns regarding data resulting from the use of AI in energy in Section 3.4.

In order to build an energy justice framework tailored to applications of AI in energy, we examine in more detail in the following sections the implications of AI for each dimension of energy justice mentioned previously in Table 1. We then propose a tailored framework in Section 3.5 and provide a demonstration of its application in Section 4, using selected use-cases of AI in the energy sector. These examples only serve to illustrate how the framework could be used and are not intended to provide specific conclusions about the technologies in question.

3.1. Distributive justice

In the context of AI in the energy system, this pillar of energy justice demands a fair allocation of the benefits and burdens of AI used for energy, throughout its whole lifecycle and across generations. It means ensuring everyone has access to clean and affordable energy, while preventing vulnerable communities from disproportionately shouldering the environmental or economic costs of new AI applications and infrastructure.

3.1.1. Impacts of additional energy infrastructure

AI services have environmental and social impacts throughout their lifecycle due to their use, associated raw material extraction, manufacturing, transport and end of life. However, the broader effects are not always considered and go beyond greenhouse gas emissions (Ligozat et al., 2022).

Increasing electricity generation alone will not supply the energy for AI; investment in infrastructure will also need to be increased and this will have impacts in the areas where these infrastructure developments are located (IEA, 2025).

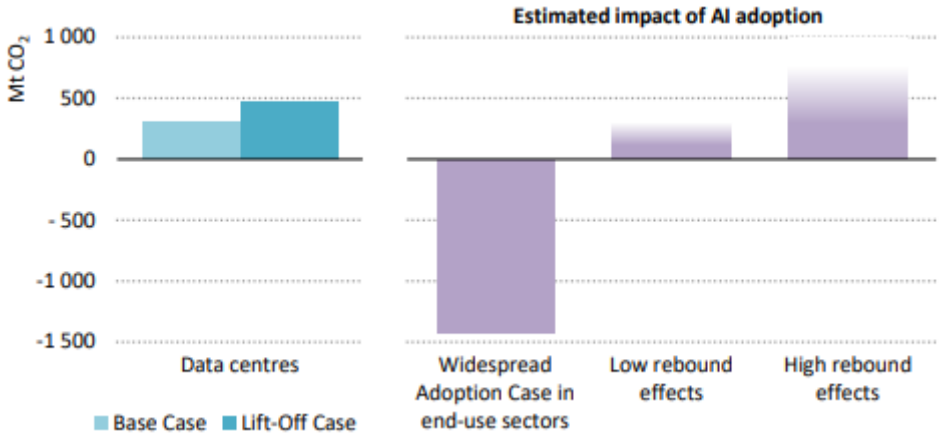
The construction of large data centres requires significant land and can put a strain on local energy grids and water resources (for cooling). This can lead to environmental justice concerns if these facilities are disproportionately located in communities that already bear a higher environmental burden. The training and operation of complex AI models also contribute to a substantial carbon footprint, potentially offsetting gains made through energy efficiency measures (Yu et al., 2024).

The IEA reports that by 2030, data centres are projected to contribute 3% of emissions from the power sector and 1% of total energy sector emissions. While the widespread use of AI applications has the potential to reduce emissions in other sectors (potentially offsetting data centres emissions by 2025) this outcome is uncertain and could be undermined by rebound effects (Figure 2), and increased fossil fuel consumption induced by AI-enabled lower prices (IEA, 2025). Furthermore, the mining, processing, and transportation of raw materials, as well as the manufacturing of AI hardware,

[10] https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:12012P/TXT&from=EN

are energy-intensive, often relying on fossil fuels, contributing to greenhouse gas emissions (van Gaalen & Chris Slootweg, 2025).

Figure 2. Indirect emissions from data centres in selected cases and an exploratory analysis of AI impacts on emissions 2025



Source: IEA, 2025

The materials required for advanced AI hardware (e.g., rare earth elements for graphics processing units (GPUs)) also have environmental and social costs associated with their extraction and disposal, often in developing countries, raising ethical questions about resource exploitation (van Gaalen & Chris Slootweg, 2025).

The most direct implication is a surge in demand for certain critical minerals and rare earth elements necessary for AI hardware, such as high-performance computing hardware, including powerful GPUs, specialized AI chips, and advanced data centers. These components rely on a complex array of raw materials, including rare earth elements used in magnets, sensors, and electronic components (IEA, 2025). Environmental impacts of raw material extraction include habitat destruction and biodiversity loss as mining operations can devastate ecosystems, leading to deforestation, soil erosion, and displacement of wildlife. Water pollution and scarcity is also a risk as mining processes are often water-intensive and can contaminate local water sources with heavy metals and toxic chemicals. The increased demand can also lead to geopolitical tensions because certain raw materials are concentrated in specific regions and competition for these resources can intensify.

This is not an exhaustive list of the lifecycle impacts of AI services. For large-scale deployments in particular, it will be crucial to evaluate their broader systemic effects because such deployments may result in societal reorganization with non-linear consequences (Ligozat et al., 2022).

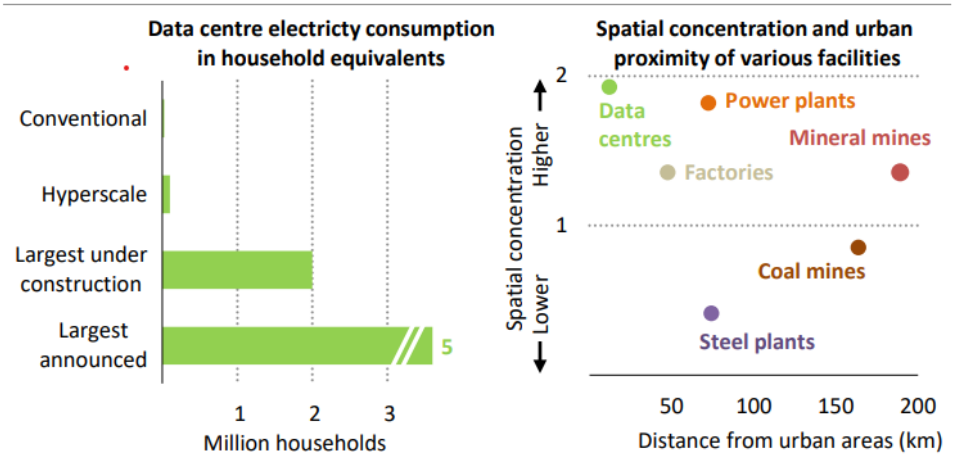
3.1.2. Affordability and Access

Affordability: the use of AI in the energy sector is predicted to optimize energy use and result in savings (IEA, 2025). It may also help to ensure reliable energy access for underserved communities. However, this depends on how it is implemented. AI could exacerbate existing inequalities if its benefits (e.g., cost savings from optimized energy use) are primarily accessible to affluent households with smart technology. If project sites or policies are selected for cost-effectiveness or efficiency only, they may result in disproportionate burdens for these communities, exacerbating existing inequalities (Delina & Tung, 2024).

Furthermore, while the application of AI for energy optimisation and decarbonisation is and will increasingly play a central role in the energy transition, the use of AI is also energy intensive in itself. Despite having energy-saving potential, AI, especially generative AI (Berthelot et al., 2024), requires significant computational power, leading to increased electricity consumption. For example, the IEA report states that, globally data centres consumed around 1.5% of electricity consumption in 2024 (IEA, 2025). The increased energy demand of AI services could result in increased pressure on energy prices in general (Bogmans et al., 2025). This latter can translate to higher energy bills for households and businesses, especially vulnerable populations, potentially leading to energy poverty.

Access: anticipating the surge in demand for AI services, investment in data centers is growing and their size is increasing. While currently handling various workloads, a typical data center might draw 10-25 megawatts (MW) of power. However, an AI-focused hyperscale data center can have a capacity of 100 MW or more, consuming as much electricity annually as 100,000 households (Figure 3)(IEA, 2025). These AI-focused data centers are expanding to accommodate ever-larger models and the rising demand for AI services. Historically, data centers have been geographically concentrated, which, given their substantial power consumption, presents considerable challenges to local power grids (de Vries-Gao, 2025) (Figure 3). The high demands of AI data centres for reliable power could lead to energy resources being prioritized for these facilities. This could potentially compromise fair energy access or the quality of supply in other areas, especially rural or currently underserved regions (Delina & Tung, 2024).

Figure 3. Data centre annual electricity consumption in household electricity consumption equivalents and the spatial concentration of various facilities versus proximity to urban areas[11]



Source: IEA, 2025

Furthermore, smart grid coverage often doesn't extend to vulnerable communities, worsening existing inequalities. In such cases, AI systems might fail to recognize the economic and geographic barriers

[11] Data centres tend to be geographically concentrated and located around cities; a 100 MW data centre can consumer as much electricity as 100 000 households (IEA, 2025).

preventing underprivileged households from accessing these systems. This oversight could lead AI to misinterpret related policies as universally beneficial for energy transition (Delina & Tung, 2024).

The adoption of smart energy technologies could also create a 'digital divide,' where those with access to technology and digital literacy benefit, while those without are left behind. This could lead to increased energy burdens for low-income households, who may not be able to afford or utilize these solutions (IEA, 2025).

3.2. Procedural justice

In the context of AI in the energy system, this pillar focuses on fairness in process. The development and governance of AI in energy must be transparent and inclusive, while ensuring the human control of AI systems. This requires clear accountability and meaningful public participation in decision-making including the design of algorithms, and accessible ways for people to seek redress from negative impacts.

3.2.1. Participation and engagement

Limited participation or failure to include stakeholders, especially underserved, disadvantaged, marginalized or vulnerable groups, such as indigenous peoples, in decisions about AI-powered energy transitions can lead to procedural injustices. When AI models are developed without open and inclusive processes that genuinely involve all stakeholders, they risk reinforcing existing power imbalances and ignoring the concerns of marginalized communities. This exclusion can lead to decisions that fail to meet their needs or represent their interests (Delina & Tung, 2024).

Furthermore, certain social classes with certain values and attitudes may be more interested in participating in the energy transition. E.g. for energy communities, things like environmental concern, desire for energy independence, trust, higher education, and awareness about local energy initiatives tend to drive participation (Thunshirn & Mlinaric, 2023). Other social groups with different values or attitudes may be left behind as a result.

It is important to note however that understandings of what 'participatory' or 'inclusive' procedures should look like may differ. Participation can be viewed as a continuum with varying degrees of decision-making power being given to participants (Arnstein, 2019). Different types of participation may be appropriate depending on the type of policy in question (Fung, 2006).

3.2.2. Transparency

AI-driven energy systems can be 'black boxes', making it difficult to understand how decisions are made. As well as this, the complexity of some AI algorithms can make it difficult to understand how they arrive at decisions, leading to a lack of transparency and accountability. Procedural justice can be undermined if marginalized communities are excluded from shaping AI algorithms and their applications. Those without the technological literacy to utilize AI driven energy solutions could be left behind.

For example, direct load control programs, while beneficial for grid management, face significant hurdles in consumer adoption due to eroding trust. This erosion stems from concerns about e.g. reduced privacy and autonomy, opaque and less predictable dynamic pricing or automation schedules, especially among individuals already worried about affordability. This perceived lack of control over utility-managed load-shifting reduces participation, as a result of mistrust and the feeling of privacy intrusion (Thunshirn & Mlinaric, 2023).

Furthermore, transparency is impacted by the outsourcing of AI development to private enterprises (Noorman et al., 2023). With the push to remove barriers for start-ups in the EU[12], including those that scale up AI innovation, it will be particularly important to ensure the transparency of AI services.

3.2.3. Human control and autonomy

Over-reliance on AI could create a situation where human operators lose the ability to effectively manage the energy system in the event of AI failures or disruptions. The complexity and rapid operational speed of autonomous AI-powered energy systems can impede the ability of human operators to intervene effectively.

For example, having AI-powered autonomously decentralized price negotiations could mean that competition and energy regulators will face considerable difficulty monitoring price trends and have limited means to control them (Noorman et al., 2023). Reliance on AI could also reduce control and autonomy for individual households (Milchram et al., 2020). This could lead to a sense of disempowerment, particularly if people feel they have no say in how AI algorithms are shaping their energy use.

3.3. Recognition justice

In the context of AI in the energy sector, this pillar is about respect and acknowledgment. It requires that AI energy solutions recognize the diverse needs, cultures, and vulnerabilities of all communities. It is about actively preventing AI from reinforcing old inequalities or creating new ones by overlooking the unique situations of marginalized groups.

3.3.1. Acknowledgement of needs

Failing to address the unique needs of vulnerable groups causes recognition injustice. Previous studies confirm that these communities already face systemic barriers in energy transition policy, hindering their adequate recognition (Delina & Tung, 2024).

AI-driven smart grids and smart home energy management systems are often designed with a 'typical' user in mind, assuming a certain level of digital literacy, access to technology, and financial means. For instance: low-income households, elderly individuals, or communities with limited internet access might be excluded from participating in or benefiting from AI-optimized demand response programs or personalized energy efficiency recommendations. Their traditional energy consumption patterns or existing coping mechanisms for energy costs may not be recognized as valid or could be simply unseen by the AI-driven system, leading to their further marginalization or increased energy burden (Tarasova & Rohracher, 2023).

While AI can identify energy consumption patterns, it might fail to recognize the complex social and economic factors that contribute to energy poverty in certain households or regions. Instead of identifying underlying vulnerabilities, it might simply categorize certain consumption as 'inefficient' or 'abnormal'. For instance, an AI system designed to optimize household energy consumption might recommend drastic reductions in heating or cooling to a low-income household struggling with energy

[12] https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/jobs-and-economy/eu-startup-and-scaleup-strategy_en

bills, without recognizing that these 'inefficient' patterns are a result of inadequate insulation, old appliances, or the need to keep vulnerable family members warm. The AI's focus on technical optimization may overlook the human stories and systemic issues contributing to energy hardship (Delina & Tung, 2024).

3.3.2. Diverse perspectives

Incorporating diverse perspectives and worldviews is particularly relevant in complex domains like energy, where solutions must be tailored to diverse societal needs and contexts, otherwise there is a risk of recognition injustice.

AI models may prioritize quantitative data and technical efficiency, potentially overlooking valuable qualitative knowledge and the lived experiences of communities regarding their energy needs, environmental concerns, and cultural practices (Werder et al., 2024). For example, prevailing social and moral values such as privacy, security or justice, were found to conflict with the design of smart grids in the Netherlands and the UK, showing the needs for energy justice concerns to be considered in future designs (Milchram et al., 2018).

Furthermore, regarding the location of renewable energy projects and/or data centres needed for AI services, developers might prioritize technical optimisation over the historical, cultural, or ecological significance of certain land areas to indigenous communities or long-standing local populations (Amato, 2025). Their perspectives on land use, ecological impact, or community well-being may not be respected, leading to a sense of disrespect and dispossession.

3.4. Data justice

How data is collected and used by AI in energy has various implications for each of the different dimensions of energy justice, which make it a cross-cutting issue (Chen et al., 2024). Data justice is a concept that has been developed to address the ethical and political dimensions of data collection, analysis, and use, particularly concerning its impact on underserved, disadvantaged, marginalized and vulnerable populations. It recognizes that data is not neutral and that its application can perpetuate or exacerbate existing inequalities (Taylor, 2017).

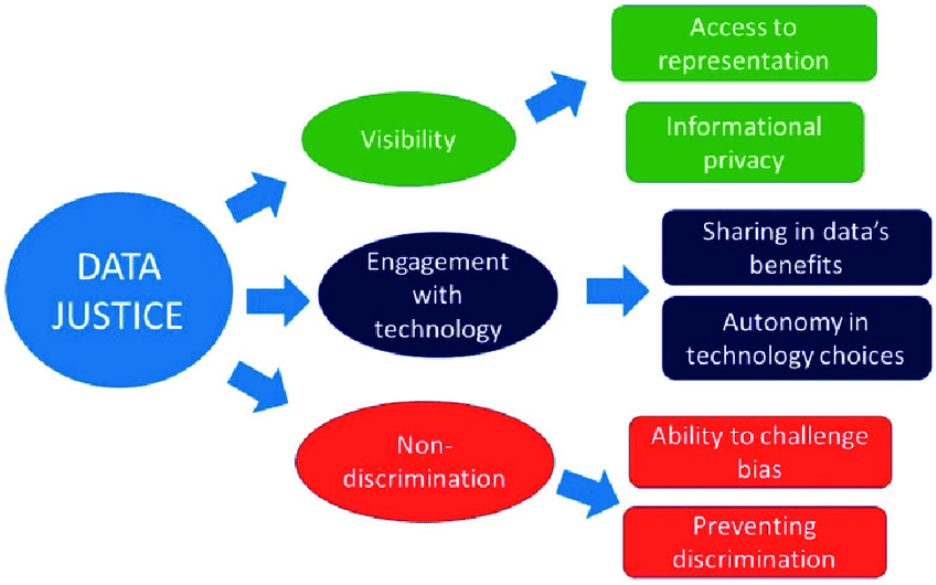
Data justice can help to ensure the ethical and fair handling of the very data upon which AI systems operate, enabling the other pillars to be truly achieved. By incorporating it into a framework for energy justice relating to AI, data justice:

- Underpins distributional justice: data justice ensures that the data inputs used by AI are fair and representative, preventing the AI from generating outputs that lead to inequitable distribution of energy burdens or benefits.
- Empowers procedural justice: data justice provides the transparency and control over data practices necessary for inclusive decision-making. While procedural justice focuses on fair processes, data justice ensures that the data informing those processes is understood, accessible, and subject to oversight by all stakeholders.
- Enables recognition justice: data justice is critical for ensuring that AI systems can recognize diverse needs by ensuring that the data they are trained on accurately and respectfully represents all communities. Without just data practices that capture true diversity, AI cannot effectively support the acknowledgment of varied identities required for recognition justice.

Protecting personal data is also a fundamental right, which is enshrined in the EU Charter for Fundamental Rights[13] as well as the Treaty on the Functioning of the European Union. Article 8 of the Charter provides the right for everyone to the protection of personal data concerning them. Such data must be processed fairly for specified purposes and with the consent of the person concerned or some other legitimate basis laid down by law. Furthermore, everyone has the right of access to data which has been collected concerning them, and the right to have it rectified.

(Taylor, 2017) offers a three-pillar framework for data justice which includes the elements of visibility, engagement with technology and non-discrimination. Visibility addresses how individuals and communities are made visible (or invisible) through data, encompassing issues of representation and informational privacy. Digital (dis)engagement focuses on the ability of individuals to engage with or disengage from data-producing technologies, ensuring autonomy and the capacity to opt out. Anti-discrimination aims to counteract and challenge data-driven discrimination, ensuring people are treated fairly based on the data collected about them.

Figure 4. Data justice framework



Source: Taylor, 2017

If that data upon which AI algorithms rely reflects existing societal biases, the algorithms will perpetuate those biases.

Distributional justice could be compromised if AI algorithms trained on biased data perpetuate discriminatory patterns in energy pricing, service provision, and resource allocation, failing to recognize the unique energy needs of vulnerable populations (Chen et al., 2024). This can perpetuate existing inequalities in access to affordable and reliable energy. For example, algorithms used to set energy prices or allocate resources may be biased against certain demographics or neighbourhoods (Delina & Tung, 2024). AI-driven energy management systems might prioritize the needs of wealthier neighbourhoods with more data available, while neglecting underserved areas (Chen et al., 2024).

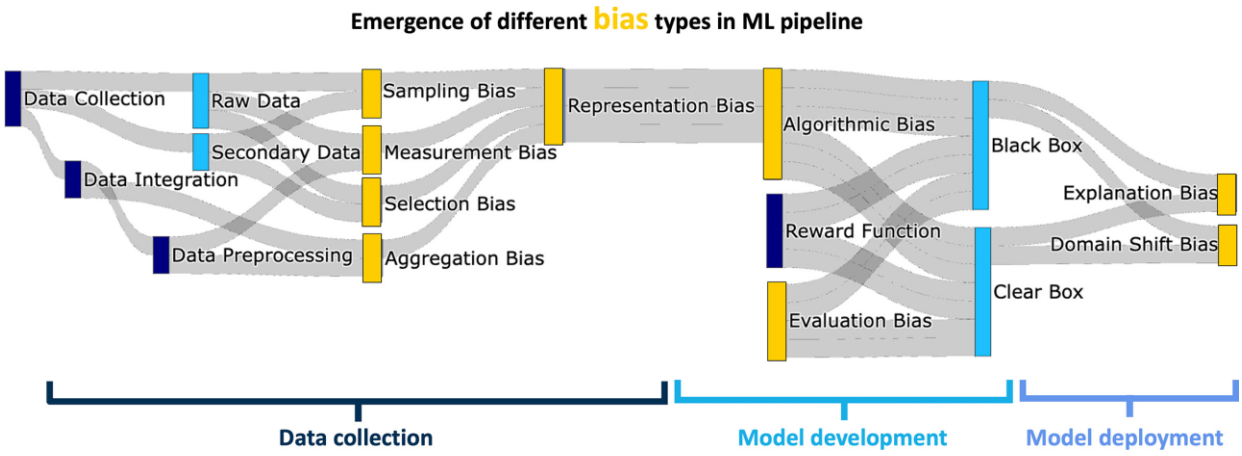
[13] https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:12012P/TXT&from=EN

Recognition justice could be compromised when training data is incomplete or biased. For example, if the data used to train AI systems doesn't accurately reflect the energy needs of the elderly and infirm, the AI may not understand their unique requirements. This can lead to exclusion and unfair treatment, violating recognition justice and worsening existing biases against vulnerable individuals and groups (Delina & Tung, 2024).

AI systems collect vast amounts of data on household energy consumption patterns, e.g. to make predictions about consumer energy demands, often without their knowledge or awareness of the implications of this. This may require personal data or data that reveals much about people's daily lives and habits (Noorman et al., 2023). Procedural justice could be compromised in this case because this data could be misused by companies or malicious actors. Vulnerable populations may be disproportionately affected by these concerns, as their data could be used to monitor and control their energy use. For example, in Spain, energy companies have been known to use data on vulnerable households to disconnect them in cases of late payments (Casals et al., 2020).

The different types of (machine learning (ML)) bias and how they impact on each of the three dimensions of energy justice (Table 2) have been described in greater detail by (Chen et al., 2024). Bias can arise during data collection, model development or model deployment(Figure 5).

Figure 5. Connecting hurdles, their origins, and their impacts throughout the ML cycle



Source : Chen et al., 2024

Table 2. Summary of types of energy justice and associated bias with examples

Type of energy justice	Potential Data and ML Bias	Examples
Recognition	Sampling Selection Representation Aggregation	Omitting to collect samples from underrepresented or marginalized groups compromises the integrity of a general population study
Distributive	Selection Measurement Algorithmic Evaluation	Collecting energy consumption data exclusively from households equipped with smart meters may lead to underestimating energy usage in households without this technology. Omitting key variables and concepts that could estimate the outcomes leads to an inaccurate assessment of unfair distribution.
Procedural	Measurement Aggregation Explanation Domain-shift	Collecting information exclusively from decision-makers while neglecting non-decision-makers' perspectives skews the understanding of the situation. Failing to gather information on critical variables essential for estimating the fairness of the decision-making process results in an incomplete analysis.

Source: Chen et al., 2024

It is therefore clear that data justice is a cross-cutting dimension of energy justice that underpins and fundamentally influences all three energy justice pillars when AI is involved. For the purposes of our framework, data justice in the context of AI in energy comprises of four main areas:

— *(In)visibility and fair representation*: ensuring data accurately portrays communities, preventing invisibility and also respecting informational privacy (invisibility) when desired (Taylor, 2017).

— *Equitable disengagement:* empowering agency over data interactions including the right to control or cease technological engagement (Taylor, 2017).

— *Bias mitigation and non-discrimination:* identifying and challenging algorithmic biases to prevent discriminatory outcomes in e.g. energy access or affordability (Chen et al., 2024; Taylor, 2017).

— *Data privacy and security:* protecting sensitive energy consumption data from misuse, breaches and unauthorized surveillance (Taylor, 2017).

3.5. Proposed framework

This section aims to present an assessment framework that is both theoretically sound and practically applicable for conducting risk assessments under the EU AI Act within the energy sector. This framework is structured to guide (risk/policy) assessments and as such, merges energy justice principles and concerns specific to AI in the energy sector.

Figure 6. Proposed energy justice framework for AI application in energy

Distributional	**Procedural**	**Recognition**
Impacts of new energy infrastructure	Participation and engagement	Acknowledgement of needs
Affordability and access	Complexity / transparency	Diversity of perspectives
	Human control and autonomy	
Data Justice		
(In)visibility		
Equitable disengagement		
Bias mitigation and non-discrimination		
Data privacy and security		

Source: JRC, 2025

Figure 6 shows the energy justice framework with four dimensions relevant to AI applications in energy. These four dimensions are:

— Distributional Justice, which refers to where injustices occur, i.e. the distribution of benefits, harms or responsibilities due to developments in the energy system and calls for (environmental) burdens and benefits to be fairly distributed across society, borders and generations and for any injustices to be rectified.

  In the context of AI applications in energy, this in particular concerns impacts of additional energy infrastructure and affordability and access (see Section 3.1),

— Procedural Justice, which refers to whether or not there is fair (due) process, that engages all stakeholders in a non-discriminatory way and calls for decision-making processes to be equitable, transparent and of high quality.

  In the context of AI applications in energy, this in particular concerns participation and engagement in the development of AI, as well as transparency regarding design and the promotion of human control and autonomy (see Section 3.2).

— Recognition Justice, which refers to: who might be ignored, i.e. the fair representation of individuals within the energy system, as well as their right to political participation taking account of divergent perspectives due to culture, gender, ethnicity, etc, and calls for: characteristics and needs of a community to be recognised, and individuals to be included, un-coerced and empowered in decision-making.

In the context of AI applications in energy, this concerns acknowledging of the needs of impacted parties, as well as the inclusion of diverse perspectives in design of AI or energy services (see Section 3.3).

— Data Justice (see Section 3.4), underpins and supports the other three dimensions above, and concerns the right to (in)visibility in data, equitable disengagement from energy-related technologies, bias mitigation and non-discrimination and data privacy and security in the energy sector.

While the framework we present is standalone, in the next section, we use in-house (JRC) examples of AI use-cases to demonstrate how the framework could potentially be applied in an energy justice assessment, bearing in mind that the results of these assessments are not conclusive and serve an illustrative purpose only.

4. Demonstrating the proposed framework

In this section, we demonstrate the use of our proposed framework using two examples of AI applications in the energy sector. We chose two studies carried out in JRC, one relating to identification of EVs for improved grid functioning and one relating to improved energy forecasting in renewable energy communities. We carry out an analysis of these studies based on available literature only. There was no contact with the participants in the studies. Since the development of our framework is a work in progress, and subject to various caveats, as we explain in the discussion (Section 5), our assessment of these two use-cases was done using immediately available data for demonstration purposes only and should not be used to make any final conclusions about the technologies in question.

4.1. Example 1: Machine learning identification of EV from charging session data

An experiment was conducted at the Joint Research Centre (JRC) Ispra campus (Ferretti & De Paola, 2025), utilizing a parking facility equipped with nine AC mono-phase wall-box chargers and a solar canopy. The facility allows for the collection of real-time power data, including drawn current and session duration.

Aim: to develop a novel method to identify Electric Vehicle (EV) attributes (brand, model, year, battery capacity) during standard AC charging sessions without hardware modifications (i.e.: the proposed machine learning approach for identifying EVs does not require any physical modifications or additions to the existing EV charging stations or the vehicles themselves).

Measurements: for each charging session, the actual drawn current in response to the excitation signal and the accumulated delay—the time between one-second sampling intervals and the latest current reading—were measured. These measurements served as the primary inputs for the machine learning model.

Data generation: the data collection involved 65 different EV models, with tests performed every 10 minutes until 20 separate testing instances were completed for each vehicle type. This resulted in two training datasets, each containing 1,300 independent entries.

Machine Learning Model: a Random Forest classifier (based on decision trees) was employed to determine key EV attributes, including: brand, model, production year, and usable battery capacity, based on the collected charging profiles.

Applications: possible applications of the model range from the design of smart charging facilities capable of identifying regular users and forecasting their charging patterns to the real-time estimation of the aggregate flexibility of connected EVs, an essential component in vehicle-to-grid (V2G) applications.

Advantages of the approach: the approach enhances the implementation of advanced smart charging strategies, support ancillary services provision to the grid, and enables real-time estimation of aggregated EV flexibility for V2G applications. From the grid's perspective:

- Enhanced grid flexibility management: precise EV model data (battery capacity, V2G potential) enables better load and resource management.
- Improved ancillary services: more accurate prediction and utilization of aggregated EV flexibility for services like frequency regulation.

— Accurate load forecasting & smart charging: model-specific data improves demand forecasting and optimizes charging schedules based on individual EV capabilities and grid conditions.

— Aggregate V2G flexibility estimation: critical for estimating total energy storage and discharge capabilities from EV fleets, vital for grid balancing.

4.1.1. Energy justice implications

Distributional justice

(a) Impacts of new energy infrastructure

Accurate forecasting of EV charging load could minimize the need for expensive grid upgrades. As well as this, understanding aggregate EV flexibility enables better integration of renewable energy (like solar and wind), helping to reduce reliance on fossil fuels.

The benefits derived from smart charging and ancillary services like V2G (e.g., grid stability, potential financial incentives for EV owners who provide flexibility) might disproportionately accrue to certain segments of EV owners. However, it should be noted that in this case, this approach enables smart features in the most common type of EV charging infrastructure (AC charging), not just more advanced types, hence helping to alleviate distributional issues. If such an approach only included more advanced types of EV charging, this could exacerbate the already uneven distribution of EV infrastructure.

(b) Affordability and access

Knowing EV specifics could help utilities improve grid stability since they can predict when and how much energy different EVs will draw or can defer, enabling smarter, more equitable demand response programs. This could prevent overloading certain areas (often low-income or marginalized communities) by optimizing charging times and avoiding unfairly shifting costs or outages onto them.

In this case, the system can lower energy costs for users who can shift consumption but excludes those with inflexible usage patterns or no access to smart devices. Thus, affordability gains may be uneven and access can be limited. Furthermore, the implementation of these sophisticated identification systems could entail costs (e.g., for data processing, integration with grid systems, potentially for upgraded charging infrastructure). If these costs are ultimately passed on to consumers, or if access to smart charging programs becomes dependent on having certain technology or vehicle types, it could place an unfair burden on lower-income individuals or communities.

Procedural justice

(a) Participation and engagement

In this case, consumers appear to have minimal participation since the system largely automates smart charging without involving users in decisions or system design. Also, the research is conducted by a research centre. There is no mention of engagement with diverse stakeholders, including vulnerable communities, low-income EV owners, consumer advocacy groups, or those potentially most impacted by new charging policies or technologies. Decisions about how and why AI is incorporated into energy infrastructure should ideally include not only AI experts but also the broad public to ensure fairness.

(b) Complexity and transparency

While the use of a Random Forest classifier may be a good technical choice, the decision-making process for identifying EVs and subsequently categorizing them for smart charging purposes might lack transparency to the end-user. EV owners might not understand why their vehicle is being categorized in a certain way and what this implies for them or for the grid, e.g. in this case enhancing grid flexibility and providing ancillary services. A lack of transparency can lead to distrust and a feeling of being unfairly treated.

(c) Human control and autonomy

The core of this technology is to extract information about the EV and its charging status without explicit input from the user or the vehicle's advanced communication systems. This means that decisions regarding charging optimization, load management, and potentially V2G interactions could be made automatically by the charging infrastructure based on inferred vehicle data, rather than direct user preferences or settings. While this can streamline processes and improve efficiency, it inherently reduces the need for direct human intervention in these aspects of the charging process.

The research focuses on the technical feasibility and accuracy of the identification method. No information is provided about the user interface or how this inferred information would be communicated back to the EV owner. A lack of transparency about how the charging system is identifying their vehicle based on inferred data could lead to a feeling of reduced control or understanding over their vehicle's behavior. However, conversely, providing this information to users could enhance their autonomy by giving them more detailed insights into their charging sessions and allowing them to make more informed decisions or customize preferences within a smart charging framework.

The degree to which this impacts human autonomy will depend on the design of the overarching smart charging systems and the extent to which user preferences and overrides are incorporated into these automated processes.

Recognition justice

(a) Acknowledgment of needs

By recognizing diverse EV capabilities, utilities could design programs accessible to a broad spectrum of EV owners, including those with older or lower-capacity vehicle. This kind of inclusivity could help to ensure all EV owners can benefit financially from smart charging incentives, not just those with the latest models.

In this case, the focus on ‘top-selling’ EV models for performance metrics, while understandable from a market perspective, risks implicitly de-prioritizing or not recognizing the needs and characteristics of other EV models or owners. This could lead to a tiered system where the charging experience and benefits for some groups are superior to others, based solely on their vehicle type. If the system primarily optimizes for, or provides better services to, newer, higher-capacity, or ‘top-selling’ EV models (as highlighted by the model's higher accuracy for these), owners of older, less common, or lower-cost EVs might be excluded from these benefits or receive fewer advantages.

(b) Diversity of perspectives

Different EV owners have diverse needs and priorities for charging (e.g., speed, cost, convenience, grid support). If the system's optimization primarily caters to grid needs or the characteristics of the most easily identifiable vehicles, it might fail to recognize and accommodate the varied needs of a diverse EV user base.

Data justice

(a) Right to (in)visibility

While necessary for this approach, continuous data collection via smart charging may compromise consumer privacy, with limited opt-out options, reducing the right to invisibility without service loss.

(b) Equitable disengagement from technologies

Those who wish to disengage from the system may risk losing access to optimized pricing or services, creating inequities for those wishing to avoid data collection or AI management.

(c) Bias mitigation and non-discrimination

While recognizing diverse EV capabilities could help tailor programs to a diversity of EV owners, there could be potential bias if AI training data includes only certain usage patterns or demographics, risking discrimination. This could be of greater concern if the project was implemented on a larger scale.

(d) Data privacy and security

Aggregated, detailed data could help utilities avoid one-size-fits-all policies that might inadvertently exclude or penalize certain EV owners or neighbourhoods. The system has the ability to identify brand, model, year of manufacturing, and usable battery capacity from charging session data. If this data can be linked to individuals (e.g., through billing information) and is collected and used without clear, transparent, and robust consent mechanisms, it can undermine trust and autonomy, impacting the recognition of individual rights.

4.2. Example 2: an open-source IT tool for energy forecast of renewable energy communities

JRC Smart Grid Lab[14] has developed an open-source IT tool designed to forecast the energy performance of Renewable Energy Communities (RECs), emphasizing accessibility and ease of use for public and community engagement (De Paola et al., 2025).

Aim: to provide an analysis of RECs of different types and compositions, assessing representative technical and economic metrics of REC performance. The developed software can estimate the hourly aggregate profiles of the analysed REC in terms of power load, generation and self-consumed energy within the REC members.

The Web-Platform includes three models:

[14] https://joint-research-centre.ec.europa.eu/laboratories-z/smart-grid-interoperability-laboratory_en

— User interface: enabling users to input parameters, select models (IRSM or GTM), and view results through a four-step web interface.

— API services: Acts as a bridge between the user interface and simulator

— Simulator module: executes forecasts using Python, FastAPI and Celery

Forecasting Methods:

— Iterated Random Sampling Model[15] (IRSM): uses historical data and Monte Carlo sampling to predict REC performance, clustering users by type (prosumers/consumers) and simulating scenarios based on geographical and weather data

— Georeferenced Temporal Model[16] (GTM): employs long-short term memory (LSTM) neural networks for forecasting, offering improved accuracy compared to IRSM. Results include metrics like aggregate load, PV production, and self-consumption.

Applications: the tool is applied to a case study, a REC in Konstanz, with 20 prosumers and 30 consumers. The tool estimates metrics like self-consumption rates and energy costs.

Summary: the tool provides techno-economic metrics (e.g., self-consumption rates, energy costs) and relies on AI-driven analysis of historical and real-time data, combining geospatial, temporal, and user-specific features. AI is central to the forecasting methodologies, platform architecture and data science, all this enables advanced predictive analytics for RECs. The tool aims to aid long-term REC planning by providing open collaboration opportunities and tailored forecasts for diverse community compositions.

4.2.1. Energy justice implications

Distributional justice

(a) Impacts of new energy infrastructure

Community energy systems powered by AI require investments in local renewables and storage infrastructure. While promoting clean energy locally, initial costs and infrastructure setup could exclude marginalized or resource-poor communities.

(b) Affordability and access

The open-source, web-based nature of the AI tool 'democratizes' access to advanced forecasting capabilities, potentially empowering smaller or under-resourced communities to plan RECs effectively and also lowering long-term energy costs and improving affordability for participants. However, AI may also magnify the digital divide, since it relies heavily on certain infrastructure and skills. While this divide is not caused by AI itself, communities with better digital infrastructure, data availability, or technical literacy may benefit more from the tool, potentially widening the gap between urban and

[15] IRSM employs probabilistic sampling and scenario generation which are aligned with data-driven AI approaches for forecasting

[16] GTM leverages machine learning libraries for predictive modelling of energy demand and renewable generation profiles

rural or affluent and marginalized communities. Accurate forecasting can help optimize self-consumption and reduce energy costs, but only for those who can implement and act on the insights.

Procedural justice

(a) Participation and engagement

There appears to be a strong emphasis on involving community members in AI design, decisions, and energy management, fostering participatory governance models.

(b) Complexity and transparency

According to the researchers, transparency and explainability are prioritized, with efforts to make AI decisions understandable to community members. The tool's open-source nature supports transparency and collaborative development, which aligns with procedural justice by allowing scrutiny and adaptation by diverse stakeholders. The graphical interface is designed for non-technical users, which helps lower barriers to participation in energy planning processes.

Still, the use of LSTM neural networks introduces complexity that may not be easily understood by non-experts. While the tool is designed to be user-friendly, the underlying AI remains a "black box" to many, which could limit meaningful participation in energy planning.

(c) Human control and autonomy

The research supports human-in-the-loop decision-making at the community level, enhancing user control and collective autonomy. However, in general, while AI can empower communities, over-reliance on opaque models may reduce local agency in decision-making.

Recognition justice

(a) Acknowledgment of needs

The tool does techno-economic forecasting with no inclusions of socio-economic data (e.g. gender, inequalities) and social considerations. It allows users to input local data or use standard profiles, which can help tailor forecasts to specific community needs. However, it may still overlook socio-cultural energy practices or informal energy use patterns.

However, the authors mention plans to integrate socio-economic metrics (e.g., household size, income, habits), which could enhance recognition justice by better reflecting the lived realities of diverse community members.

(b) Diversity of perspectives

The research seeks to integrate multiple stakeholder perspectives through participatory design, meaning that there is a greater chance for marginalized voices to influence AI design and energy planning.

Data justice

(a) Right to (in)visibility

Communities without historical energy data or digital records may be underrepresented in the model's training data, leading to less accurate forecasts and potential misrecognition of their energy behaviours. On the other hand, the tool uses primarily operational data and personal user data is not collected, so invisibility concerns are minimal.

(b) Equitable disengagement from technologies

The focus on community governance may enable more equitable options to disengage or participate, respecting member autonomy.

(c) Bias mitigation and non-discrimination

The research emphasises inclusive datasets and bias mitigation to ensure fair AI performance across diverse community members. However, if the training data used in the LSTM model is skewed toward certain regions or user types, it could introduce bias in forecasts, affecting the fairness of outcomes.

(d) Data privacy and security

The authors primarily focus on the technical functionality and accuracy of the forecast tool, with less detail about privacy and security safeguards.

The GTM requires various parameters for each REC member, including user type and category, geo-localisation, contracted power, etc. Users can upload their historical energy consumption and generation data. This granular data can reveal daily routines, presence at home, and usage patterns of appliances, making it highly sensitive. The degree to which individual user data is anonymized before processing and aggregation is crucial for privacy. If raw, identifiable data is stored or processed, even temporarily, it poses a privacy risk.

Being open source allows for transparency and community auditing, which can be beneficial for identifying and fixing vulnerabilities. However, it also means the code is publicly available, which could potentially expose architectural details that malicious actors could exploit if not properly secured. Various aspects of the tool' s architecture are subject to security risk e.g. the API, the Redis database, data transmission or authentication or authorization aspects.

The collection of detailed user data, its aggregation, and the web-based architecture would therefore require robust privacy-preserving measures (e.g., strong anonymization techniques, data minimization, clear consent) and comprehensive security protocols (e.g., secure APIs, encrypted data transmission, robust access controls, secure database configuration) to protect sensitive information and maintain user trust.

5. Discussion

In this report we propose a guidance framework for assessing energy justice implications of AI used in energy, since no such framework exists at EU level. Before applying the framework, however, several issues should be taken into consideration.

Firstly, users should be mindful that any such framework will always be based on a particular set of normative assumptions, and these should be made transparent. Since it is based on current energy justice theory as well as existing general ethical guidelines for AI, we consider this framework to be a good starting point. However, the interpretation of normative premises in the framework may differ depending on who is applying it, and this can change resulting policy recommendations. Following the recommendations of (Van Uffelen et al., 2024), we advise that the elements of the framework (i.e. distributional, procedural, recognition and data justice) be used as categories only and that interpretations of these categories be made according to the AI application in question. We provided some examples of potential ways of applying the framework in this way in Section 4, where we analyse two different AI applications. We stress that these are illustrative examples only and the results are not conclusive with regard to the technologies in question. We did an analysis based on immediately available data from the literature. However, ideally, a more thorough qualitative approach could be taken where analysis of the use-cases in question could involve discursive methods, in-depth interviews, surveys or other qualitative approaches to gather more information from participants of the studies, which could be analysed according to the categories of our proposed framework.

Secondly, users should be mindful of the subject of justice (Van Uffelen et al., 2024) of the framework, i.e. to whom should the principles apply? Thirdly, the scale of justice is also relevant – for different applications of AI, the scale of implementation will result in different orders of impact, some of which may be global. Supply chains for infrastructure and equipment, even for small scale projects may still have global impacts.

Furthermore, users should also be mindful of the rapidly changing regulatory landscape around AI. What may be considered acceptable today may not be acceptable in the near future, or vice versa. In this regard, the timescale of any assessment is also important to consider. Our guidance framework should therefore be considered as non-exhaustive and open to continuous change.

It would therefore benefit from the inputs of a multi-stakeholder forum, e.g. involving policymakers, industry, civil society, consumer groups, and academics, to continuously refine, adapt, and oversee its application, and ensure that it remains relevant, responsive to emerging challenges, and grounded in diverse perspectives, promoting procedural and recognitional justice in its own governance.

Bearing the above points in mind, we see some potential uses for the framework such as:

- Integration into a conformity assessment procedure for high-risk AI systems in the energy sector, as defined by the EU AI Act, as a (mandatory) component. This would ensure that energy justice considerations (distributive, procedural, recognitional) are systematically evaluated from the design phase through deployment, proactively mitigating potential harms.
- Employment as an evaluation tool within regulatory sandboxes or pilot projects for novel AI energy technologies. This would allow testing of the framework's efficacy, identification of unforeseen justice impacts, and gathering of data for future policy adjustments before widespread deployment.

6. Conclusions

The integration of Artificial Intelligence into the energy sector holds immense potential for advancing the European Union's green and digital transition goals. However, as this report underscores, the deployment of AI in energy is not without significant energy justice implications across distributive, procedural, and recognition dimensions.

AI-driven energy systems can inadvertently exacerbate existing inequalities—especially those caused by digitalisation in general—or create new ones. Distributive injustices may arise from the environmental burdens of AI infrastructure (e.g., data centres) and raw material extraction, as well as from potential increases in energy prices and unequal access to AI-enabled energy benefits, disproportionately affecting vulnerable populations. Procedural injustices can manifest through a lack of transparency in AI decision-making, limited stakeholder participation in AI development, and the erosion of human control and autonomy over energy systems. Furthermore, recognition justice is compromised when AI systems fail to acknowledge the diverse needs, cultural practices, and vulnerabilities of marginalized groups, leading to their exclusion and unfair treatment.

The EU AI Act provides a foundational regulatory framework, particularly for high-risk AI systems in critical infrastructure like energy, by emphasizing fundamental rights, human oversight, transparency, and accountability. However, a dedicated energy justice framework tailored to AI is essential to address the unique complexities of the energy sector and ensure a truly just transition. By integrating principles of energy justice and data justice (i.e. ensuring fair data collection, usage, and representation), this framework can underpin the effective achievement of distributive, procedural, and recognition justice in AI-assisted energy transitions. Ultimately, proactively addressing these justice dimensions is critical for building public trust and ensuring that the benefits of AI in energy are equitably shared across all members of society.

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List of abbreviations and definitions

Abbreviations	Definitions
AI	Artificial Intelligence
API	Application Programming Interface
DSO	Distribution System Operator
EV	Electric Vehicle
GPU	Graphics Processing Unit
GTM	Georeferenced Temporal Model
IRSM	Iterated Random Sampling Model
JRC	Joint Research Centre
LSTM	Long-Short Term Memory
ML	Machine Learning
PV	Photovoltaic
REC	Renewable Energy Community
TSO	Transmission System Operator

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